

Experiment 'p_4_6_find_minimum_array' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_4_6_find_minimum_array

Natural language statement: Write a method that finds the minimum value from a list of inputs into a method.

Method signature: p_4_6_find_minimum_array(inputs: array<int>) returns (min: int)

Ensures

- forall i :: 0 <= i < inputs.Length ==> min <= inputs[i]
- exists i :: 0 <= i < inputs.Length && min == inputs[i]

Requires

- 0 < inputs.Length

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that finds the minimum value from a list of inputs into a method.

The signature should be:

`p_4_6_find_minimum_array(inputs: array<int>) returns (min: int)`

The method should respect the following contract:

requires `0 < inputs.Length`, ensures forall `i :: 0 <= i < inputs.Length ==> min <= inputs[i]`
↪ `i]`, ensures exists `i :: 0 <= i < inputs.Length && min == inputs[i]`

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmfdclsZ1R64SFsLoazS5fBOeEFTR

Model: gpt-5.5-2026-04-23

Created at: 1780492172

Finish reason: stop

Usage: 332 tokens in, and 494 tokens out

```
{
  min := inputs[0];
  var idx := 1;
  while idx < inputs.Length
    invariant 1 <= idx <= inputs.Length
    invariant forall i :: 0 <= i < idx ==> min <= inputs[i]
    invariant exists i :: 0 <= i < idx && min == inputs[i]
  {
    if inputs[idx] < min {
      min := inputs[idx];
    }
    idx := idx + 1;
  }
}
```

Final Program

```
method p_4_6_find_minimum_array(inputs: array<int>) returns (min: int)
```

```

    requires 0 < inputs.Length
    ensures forall i :: 0 <= i < inputs.Length ==> min <= inputs[i]
    ensures exists i :: 0 <= i < inputs.Length && min == inputs[i]
{
  min := inputs[0];
  var idx := 1;
  while idx < inputs.Length
    invariant 1 <= idx <= inputs.Length
    invariant forall i :: 0 <= i < idx ==> min <= inputs[i]
    invariant exists i :: 0 <= i < idx && min == inputs[i]
  {
    if inputs[idx] < min {
      min := inputs[idx];
    }
    idx := idx + 1;
  }
}

```

Total Token Usage

Input tokens: 332

Output tokens: 494

Reasoning tokens: 369

Sum of 'total tokens': 826

Experiment Timings

Overall Experiment started at 1780492171920, ended at 1780492187740, lasting 15820ms (15.82 seconds)

Iteration #1 started at 1780492171920, ended at 1780492187740, lasting 15820ms (15.82 seconds)